

RIVKA SORSCHER

SOFTWARE ENGINEER

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High-achieving Mathematics and Computer Science student with outstanding academic performance (GPA 92). Possesses strong analytical and mathematical abilities, along with a proactive, responsible, and goal-oriented approach. Demonstrates solid interpersonal skills and a high level of motivation. Seeking a challenging software engineering position to apply and further develop technical and problem-solving capabilities.

Work experience

2025 - present | Testing and Training International NY USA, Junior developer – student position

- Developing applications using React and PHP, working with MySQL and SQL Server databases, and handling data processing and backend logic.
- Collaborating with senior developers to design and implement integrations between the company's platforms, including development in a .NET codebase calling REST API endpoints.

Education

2026 - present | Bar Ilan University

- Master of Science, Mathematics with thesis
- Participated in the 2026 Women's Mathematics Research Students Conference hosted by the Einstein Institute of Mathematics at The Hebrew University of Jerusalem.

2022 - 2026 | Jerusalem College of Technology – Lev Academic Center

- Bachelor of Science, Computer Science
- Grade average: 92
- Relevant coursework includes: Algorithms, Operating Systems, Data Structures, Database Systems, Calculus 1 & 2, Statistics, Probability, Introduction to AI, Computer Architecture (RISC-V)

Academic projects

- Designing a Rubik's cube solving algorithm in **Python**, utilizing **artificial intelligence heuristics** for efficient solution strategies, increasing runtime performance by over 20%.
- Developing a **Java**-based ray tracing engine, including **geometric modeling**, lighting and rendering algorithms, along with performance and image quality improvements. Performance improvements including **adaptive super sampling** and **multi-threading**. Image quality improvements including **anti-aliasing** implemented based on super sampling.
- Building a website for selling second hand furniture. The site implements **token-based** authentication and authorization and integrates with Google Maps to provide location-based functionality. The site is built with **JavaScript, React, Node JS, MySQL** technologies combined with version management in **Git**. The site was designed with **web accessibility**, offering a convenient and pleasant user interface to all users.
- Developing a comprehensive **PostgreSQL** database for managing bakery operations, including branches, employees, recipes, inventory, and production. Building **ERD/DSD** models, automated data generation, and implemented advanced queries, constraints, and transaction handling. Creating **PL/pgSQL** functions, procedures, and triggers for shift scheduling, production workflows, and full **CRUD** logging.

Technical skills

Languages: Python, C++, C, Java, C#, JavaScript, Bash
Databases: Microsoft SQL Server, MySQL, PostgreSQL
Frameworks: .NET, Node.js, Web API
Web technologies: React, HTML, CSS, TypeScript
Operating systems: Linux, Windows

Spoken languages

English as mother tongue, Hebrew as mother tongue.